

A Physics-Informed Neural Network for Option Pricing under Rough Volatility: Meshfree Solution of the Markovian-Lifted PDE and Fast Parametric Calibration

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Draft: Introduction + Related Work + Methods + Results (Abstract and Discussion to follow)

Abstract

[Abstract — single structured paragraph, no citations/figures, written last (Stage 4).]

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1 Introduction

Pricing a derivative under a stochastic model amounts to solving a partial differential equation (PDE). For most models this PDE is low-dimensional and classical grid-based solvers are effective. An important and empirically well-supported class of models, however, breaks this picture: under *rough volatility* the instantaneous variance is driven by a fractional noise with Hurst exponent $H < 1/2$, so the variance process is neither Markovian nor a semimartingale and there is *no* finite-dimensional pricing PDE (Gatheral et al., 2018; El Euch and Rosenbaum, 2019). The standard remedy is the *Markovian lift*: the fractional kernel is approximated by a sum of n exponentials, restoring a PDE at the cost of raising its dimension to $n + 1$ (Abi Jaber and El Euch, 2019). Because n directly sets the dimension of the PDE while controlling a known approximation error, the lifted rough-volatility problem is a clean, economically motivated testbed for mesh-free neural PDE solvers and for studying the curse of dimensionality.

Physics-informed neural networks (PINNs) solve a PDE by minimizing its residual at sampled collocation points, using automatic differentiation and requiring no labelled solution data (Raissi et al., 2019). Being mesh-free, they sidestep the exponential growth in grid points that makes finite differences infeasible in high dimensions — the same motivation behind deep PDE solvers such as the deep Galerkin method (Sirignano and Spiliopoulos, 2018) and deep BSDE solvers (Han et al., 2018). Two properties make the approach attractive for derivatives: the physics (the PDE) is known exactly, so market data enter only through the model coefficients; and, once trained across a parameter range, the network is differentiable in those parameters, enabling gradient-based calibration.

On the finance side, rough volatility reproduces the observed roughness of realized variance and the term structure of at-the-money skew far better than classical stochastic-volatility models such as Heston (1993), and the rough Heston model in particular admits a semi-analytic characteristic

function (El Euch and Rosenbaum, 2019). Calibration, however, remains computationally demanding, which motivates fast neural surrogates. We use Bitcoin (BTC) options as a testbed: the market is liquid, exhibits a pronounced negative skew, and calibrates to a rough Hurst exponent.

Our contributions, stated conservatively, are the following.

1. **A dimension-scaling result (primary, methodological).** The PINN’s *solve-error* — how well it solves the PDE it is given — grows only *sub-linearly* in the lifted dimension, staying below about 11 price bp up to $n = 32$, while a naive full-tensor finite-difference grid blows up exponentially and is already infeasible by $n = 8$. We are explicit that $50^{(n+1)}$ is the *naive* grid; structured solvers do better but degrade with n and are not standard for this PDE, so the honest claim is that finite differences are *impractical*, not impossible.
2. **An error decomposition and a rule for choosing n .** We separate the *solve-error* (network versus lifted Monte Carlo at the same n) from the *lift-error* (lifted Monte Carlo versus the true model), report them in price space where the signed errors add exactly, and use their crossover to choose n ; we show how the crossover shifts with maturity.
3. **Empirical calibration.** We calibrate rough Heston to real Deribit BTC options and obtain a rough Hurst exponent with strong negative correlation.
4. **Fast parametric recalibration (applied).** A parametric PINN plus a *hybrid* (PINN warm-start followed by a few local Fourier polish steps) reaches from-scratch calibration quality at roughly nine times fewer Fourier evaluations; we report the parametric network’s standalone accuracy gap honestly.

We emphasize what we do *not* claim. We are not the first to solve the lifted (rough) Heston PDE with a mesh-free neural method: Papapantoleon and Rou (2025) already do so. Our contribution is a specific combination — a PINN residual formulation, the solve/lift decomposition and n -crossover analysis, real crypto calibration, and the parametric hybrid — detailed next.

2 Related work

Mesh-free deep pricing on the lifted PDE (closest prior art). Papapantoleon and Rou (2025) price European options in (rough) diffusion models by reformulating the pricing PDE as an energy-minimization problem and approximating it in a time-stepping fashion with deep networks (a “deep gradient flow”), demonstrated specifically on the lifted Heston model and respecting the large-moneyness asymptotics and known price bounds. This is the work closest to ours in both target (the lifted rough-vol PDE) and spirit (mesh-free deep pricing). It differs in method: theirs is a time-stepping energy/variational scheme, whereas ours is a PINN that directly minimizes the strong-form PDE residual with a hard-constrained initial condition. It also differs in scope: we add a quantitative n -scaling and solve/lift decomposition, calibration to real market data, and a parametric recalibration hybrid, none of which is the focus there.

Other neural solvers for rough volatility. Jacquier and Zurič (2023) solve the path-dependent PDEs of rough volatility by interpreting them as backward stochastic differential equations and using a reservoir-type *random* neural network, which reduces training to a least-squares regression with convergence guarantees. This shares our target class but uses a fundamentally different mechanism (BSDE plus random features, not residual minimization) and does not study dimension scaling or calibration speed, which are central here.

Neural calibration surrogates (the calibration foil). The dominant paradigm for fast (rough-)volatility calibration is the supervised surrogate of Horvath et al. (2021): a neural network is trained offline on a large grid of precomputed model prices or implied volatilities and then used as a fast pricing map inside a calibration loop. This is purely data-driven and carries no PDE consistency — the network never sees the governing equation. Our parametric network is instead *physics-informed*: it learns from the PDE residual rather than from precomputed labels. This distinction is what makes our hybrid meaningful — the physics-trained network supplies a warm start that a few true-model evaluations then refine (Section 4.4).

Foundations. We build on PINNs (Raissi et al., 2019) and, more broadly, mesh-free deep solvers for high-dimensional PDEs (Sirignano and Spiliopoulos, 2018; Han et al., 2018); on rough volatility (Gatheral et al., 2018; El Euch and Rosenbaum, 2019) and its multifactor Markovian lift (Abi Jaber and El Euch, 2019); and on classical transform pricing (Heston, 1993; Carr and Madan, 1999) for our independent ground truth. Two PINN training techniques we adopt are self-adaptive loss weighting by homoscedastic uncertainty (Kendall et al., 2018) and causal (curriculum) training in the time variable (Wang et al., 2024).

3 Methods

3.1 Rough Heston and the Markovian lift

Under rough Heston the variance is driven by the fractional kernel $K(t) = t^{\alpha-1}/\Gamma(\alpha)$ with $\alpha = H + \frac{1}{2}$ and $H < \frac{1}{2}$, which is non-Markovian (Gatheral et al., 2018; El Euch and Rosenbaum, 2019). The Markovian lift (Abi Jaber and El Euch, 2019) approximates $K(t) \approx \sum_{i=1}^n c_i e^{-x_i t}$, introducing n Ornstein–Uhlenbeck factors U^i with $V = V_0 + \sum_i c_i U^i$. The discounted price $P(t, x, u)$, with $x = \log S$, then solves the $(n+1)$ -dimensional backward Kolmogorov PDE

$$\begin{aligned} 0 = & P_t + \left(r - \frac{1}{2}V\right)P_x + \frac{1}{2}V P_{xx} + \sum_i \left[-x_i u_i + \lambda(\theta - V)\right]P_{u_i} \\ & + \frac{1}{2}\nu^2 V \sum_{i,j} P_{u_i u_j} + \rho\nu V \sum_i P_{xu_i} - rP, \end{aligned} \quad (1)$$

with terminal condition $P(T, x, u) = \text{payoff}(e^x)$ and $V = V_0 + \sum_i c_i u_i$. All factors share one Brownian motion, hence the full double sum in the u -diffusion and the cross term with x . The parameter n is thus a tunable knob on the PDE dimension. We build (c_i, x_i) by the geometric-quadrature rule of Abi Jaber and El Euch (2019); the relative L^2 kernel error decreases monotonically from 25.6% at $n = 5$ to 0.39% at $n = 100$.

3.2 Two independent ground truths

We validate against two mutually independent pricers. (i) A semi-analytic *Fourier* pricer solves the fractional Riccati equation for the rough Heston characteristic function (El Euch and Rosenbaum, 2019) and inverts it by Fourier integration (Carr and Madan, 1999); it does not use the lift, so it is the ground truth of the true model ($n \rightarrow \infty$). In the flat-volatility limit it matches Black–Scholes to 3.7×10^{-7} . (ii) A *lifted Monte Carlo* simulator integrates the n -factor lifted SDE with an exponential-integrator Euler scheme that treats the stiff $-x_i U^i$ term exactly — the mean reversions span roughly ten orders of magnitude for small H , so explicit schemes diverge. It matches Black–Scholes within Monte Carlo error. Because the two pricers approximate different objects (an ODE

versus the model dynamics), their agreement is a genuine cross-check. Crucially, *at finite n the lifted MC is the ground truth of the PDE the network solves*, whereas Fourier is the ground truth of the model; this underlies the decomposition in Section 3.4.

3.3 Physics-informed neural network

We solve (1) with a PINN (Raissi et al., 2019), a mesh-free residual method in the spirit of Sirignano and Spiliopoulos (2018); Han et al. (2018). The network is a tanh multilayer perceptron of (τ, x, u) with $\tau = T - t$, so the terminal condition becomes an initial condition at $\tau = 0$. We hard-wire that condition through the ansatz

$$P(\tau, x, u) = P_{\text{base}}(\tau, x) + \tau N_{\theta}(\tau, x, u), \quad (2)$$

whose τ prefactor forces the learned correction N_{θ} to vanish at $\tau = 0$. The baseline P_{base} is a Black–Scholes price anchored not at the spot variance V_0 but at the model’s expected *integrated* variance, $w(\tau) = \int_0^{\tau} \mathbb{E}[V_s] ds$, which captures mean reversion of V from V_0 toward θ ; anchoring at V_0 leaves a systematic level bias that the network would have to undo.

The loss combines the interior residual of (1) with asymptotic boundary losses $P \rightarrow 0$ as $S \rightarrow 0$ and $P \rightarrow S - Ke^{-r\tau}$ as $S \rightarrow \infty$, resampled each iteration. Two stabilizers matter. First, the residual and the two boundary terms are combined with self-adaptive per-term weights via the learnable homoscedastic-uncertainty scheme of Kendall et al. (2018), removing hand-tuned weights. Second, a τ -curriculum trains short maturities first and expands the horizon, respecting temporal causality (Wang et al., 2024). Finally, factor collocation is informed by a short pre-flight Monte Carlo: rather than sampling u in an arbitrary symmetric box, we sample each factor from the mean and standard deviation it actually attains under the lifted dynamics, which is essential for reproducing the skew.

3.4 Validation protocol and error decomposition

The network is trained at fixed strike $K = 1$ and prices as a function of spot; the fixed-spot smile $C(1, K)$ is recovered by the call’s degree-one homogeneity,

$$C(1, K) = K C(1/K, 1), \quad (3)$$

which holds because the variance dynamics are independent of the spot level. Omitting this rescaling compares mismatched options and manufactures thousands of basis points of spurious error (here ~ 2700 vol bp reduced to ~ 286 once corrected). We never report a lone “PINN-versus-Fourier” number; instead we decompose, per strike and in price space where the signed errors add exactly,

$$\underbrace{(\text{PINN} - \text{Fourier})}_{\text{total}} = \underbrace{(\text{PINN} - \text{MC})}_{\text{solve-error}} + \underbrace{(\text{MC} - \text{Fourier})}_{\text{lift-error}}. \quad (4)$$

The solve-error is what training, architecture, and sampling can improve; the lift-error is a model approximation independent of the network, shrinking only with n . We report price RMSE in basis points of spot (where (4) is additive) and give implied-volatility equivalents where useful; the two components have opposite signs and so do not add in RMS.

3.5 Parametric PINN

For instant recalibration we train one network that also takes the five model parameters $(H, \nu, \rho, \lambda, \theta)$ as inputs, over a box around the calibrated BTC values $(H \in [0.04, 0.20], \nu \in [0.10, 0.60], \rho \in$

$[-0.95, -0.20]$, $\lambda \in [0.5, 4.0]$, $\theta \in [0.08, 0.35]$; V_0 and the maturity are fixed. Since the lift weights depend on H , they are precomputed on an H -grid and gathered per collocation point, while the parameters enter the network as exact continuous inputs. The number of factors is fixed by the crossover rule of Section 4.1 ($n = 10$). The network is differentiable in the model parameters, which enables gradient calibration.

3.6 Calibration procedures and data

We calibrate three ways: (a) *Fourier-in-the-loop*, differential evolution with the Fourier pricer inside the objective; (b) *PINN gradient*, Adam on the parameters using the parametric network as a differentiable pricer, with zero Fourier calls; and (c) *hybrid*, the PINN’s single-shot solution as an initial guess refined by a few local Fourier polish steps (Nelder–Mead). The objective is the vega-weighted implied-volatility RMSE against the market smile. The market data are end-of-day BTC option quotes from Deribit, a cryptocurrency derivatives exchange and data provider, retrieved through its public interface; we set $r = 0$ and infer the forward from the quotes, following the crypto convention, and recompute implied volatilities from mid prices with a robust Black–Scholes inverter.

4 Results

4.1 Dimension scaling: sub-linear solve-error versus FD blow-up

Training the same architecture (width 64, depth 4) at $n = 4, 8, 16, 32$ on the canonical BTC calibration at $T = 0.15$ and measuring the solve-error against the lifted MC over three seeds (Table 1, Figure 1), the solve-error grows only mildly — from 5.4 ± 1.0 to 10.7 ± 2.0 price bp, about a factor of two over an eight-fold increase in dimension. It is not flat (an earlier single-seed run misleadingly suggested so), but grows far more slowly than the dimension. Wall-clock cost to the plateau stays bounded (about 6–10 minutes on a CPU; per-iteration cost roughly constant at 55–62 ms) because the factor Hessian is vectorized.

By contrast, a naive full-tensor finite-difference grid on the same $(n+1)$ -dimensional PDE at 50 nodes per dimension needs $50^{(n+1)}$ points: 3.1×10^8 at $n = 4$, 2.0×10^{15} at $n = 8$, and 1.2×10^{56} at $n = 32$. Even $n = 8$ (about 16 petabytes in double precision) is infeasible, while the network trains every n on a laptop CPU. We state the contrast honestly: $50^{(n+1)}$ is the *naive* grid. Structured schemes — sparse grids, ADI, low-rank tensor-train — push the tractable dimension higher, but their cost and accuracy still degrade with n (sparse-grid node counts grow super-linearly; tensor-train needs low-rank structure that the coupled ρ – ν cross terms erode), and none is standard or validated for the lifted rough-volatility PDE. The honest claim is that a competitive finite-difference solve at $n = 16$ – 32 is *impractical*, not provably impossible. Gentle solve-error growth with bounded cost, against this blow-up, is the curse-of-dimensionality argument for the mesh-free approach.

4.2 The solve/lift decomposition and a crossover rule for n

The decomposition explains the record. A single “PINN vs Fourier” implied-vol RMSE of ~ 114 – 127 vol bp at $n = 4$ would read as network error; split by (4), it is dominated by the lift. In the full configuration the $n = 4$ error is 37 vol bp of solve-error plus 133 of lift (partially cancelling to a total of 121), and in the correlation-free case it is only 8 of solve-error on top of 133 of lift. The network already solves its PDE to well under 50 vol bp; the apparent gap was model approximation.

Table 1: Dimension scaling (canonical BTC calibration, $T = 0.15$; solve-error is 3-seed mean $\pm$ SD; price RMSE in basis points of spot).

n	solve-error (px bp)	lift-error (px bp)	train time (s)	FD grid $50^{(n+1)}$
4	5.4 ± 1.0	16.4	397	3.1×10^8
8	6.1 ± 1.4	7.9	386	2.0×10^{15}
16	8.8 ± 1.6	3.5	591	7.6×10^{28}
32	10.7 ± 2.0	1.7	457	1.2×10^{56}

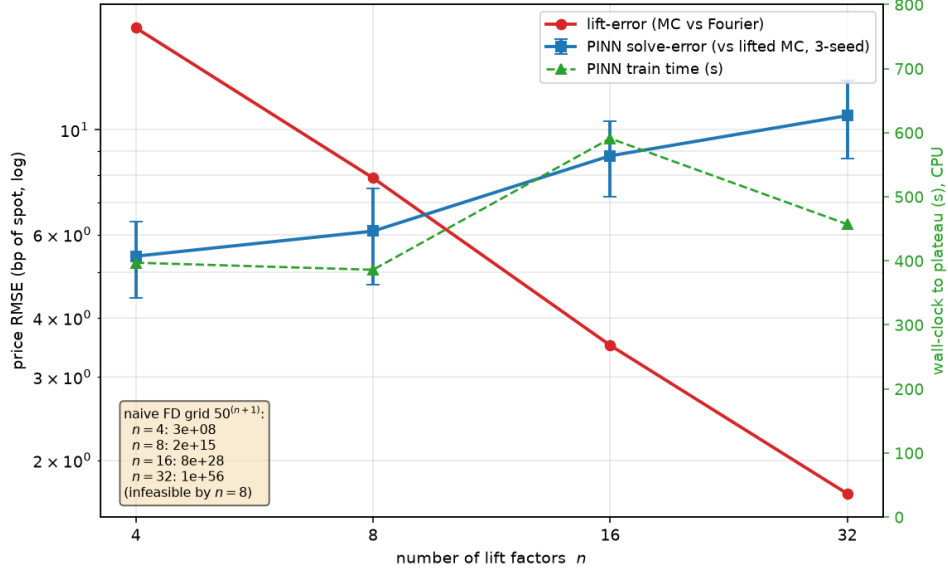


Figure 1: Dimension scaling. PINN solve-error (blue, 3-seed mean $\pm$ SD) grows sub-linearly and stays under ~ 11 price bp to $n = 32$, while lift-error (red) falls monotonically and training time (green) stays bounded; the naive finite-difference grid (inset) is infeasible by $n = 8$.

Because the lift-error shrinks monotonically (133, 63, 27, 12 vol bp at $n = 4, 8, 16, 32$; equivalently $16.4 \rightarrow 1.7$ price bp) while the solve-error grows mildly, the two cross. At $T \approx 0.15$ the crossover sits near $n = 8$ – 10 (Figure 1): below it the model approximation dominates and one should add factors; above it the network’s own error dominates and factors are wasted. This gives a concrete rule for choosing n . The crossover also moves with maturity: over a T -sweep the solve-error stays bounded everywhere (roughly 1–9.5 price bp across $T \in \{0.05, 0.15, 0.5\}$ and $H \in \{0.05, 0.10, 0.15\}$), but the lift-error grows with T — from 8.6 to 31.8 price bp at $n = 4$ as T goes $0.05 \rightarrow 0.5$ — so at long maturities the lift still exceeds the solve floor at $n = 16$ and the crossover shifts to higher n . Longer horizons need more factors.

4.3 Calibration to real BTC options gives a rough Hurst exponent

Calibrated to the Deribit BTC surface with a full-budget optimizer (differential evolution, 60 generations plus local polish; all 13 maturities, up to 8 points per maturity, 97 points), the model yields $H = 0.090$, $\rho = -0.792$, $V_0 = 0.103$, $\theta = 0.253$, $\lambda = 2.567$, $\nu = 0.300$, at a vega-weighted implied-vol RMSE of 496 vol bp (Figure 2). The Hurst exponent is well inside the rough regime

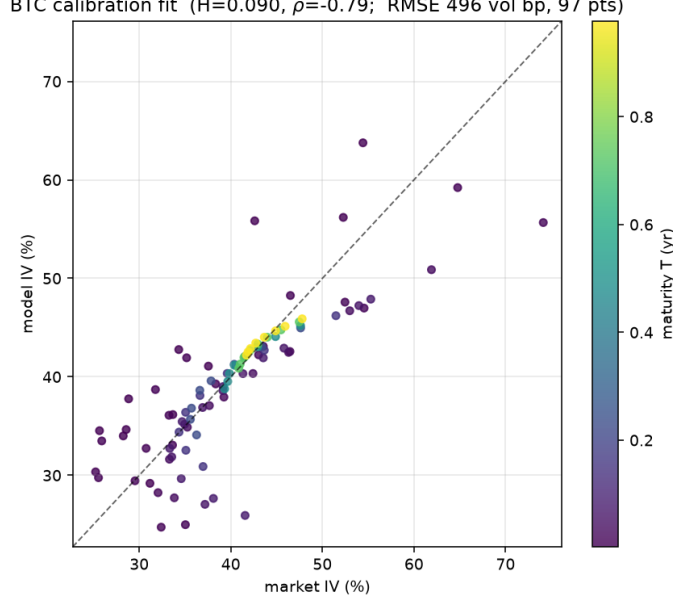


Figure 2: Rough Heston calibrated to the Deribit BTC surface (97 points, 13 maturities, colour = maturity): market versus model implied volatility at the canonical fit ($H = 0.090$, $\rho = -0.79$; RMSE 496 vol bp). The dashed line is the identity.

and the correlation strongly negative, both economically sensible for crypto. We report the fit as deliberately coarse: crypto smiles are wide and the lifted rough-Heston form is stiff, and a single 496-bp figure is honest about that.

4.4 Parametric PINN and the calibration hybrid

A single network covering the five-parameter box is necessarily less accurate at any one point than a network trained there. At the box centre the parametric solve-error is 11.0 ± 4.7 price bp against 5.9 ± 1.2 for a single-point network at the same $n = 10$ — about a two-fold coverage penalty. The penalty is uneven: it stays near 9–11 price bp at typical points but rises to 47.7 in the rough low- H corner and 29.8 at high vol-of-vol, where the PDE is hardest. We report this honestly; a larger training budget would shrink the corner penalty.

This limits standalone PINN calibration but not its use as a warm start. Calibrating the BTC $T \approx 0.15$ slice three ways (Table 2), the Fourier-in-the-loop optimizer reaches 23.7 vol bp using 2018 Fourier evaluations in about 168 s. PINN-only gradient descent is fast (about 2 s, zero Fourier calls) but poor — 222 ± 20 vol bp under the true model — because it calibrates to its own imperfect prices and, as an honest aside, because this single-maturity slice optimum lies outside the parametric training box (it wants $\nu \approx 0.87$, $\theta \approx 0.39$). The hybrid resolves this: seeding the local Fourier polish with the PINN’s guess reaches 29.4 ± 7.1 vol bp — comparable to from-scratch on a 34–52% smile — in only 237 ± 46 Fourier evaluations and about 23 s, roughly nine times fewer Fourier calls and seven times faster. The applied value of the parametric network is therefore as a calibration warm start, not a standalone pricer.

Table 2: Calibration of the BTC $T \approx 0.15$ slice, three ways (3 PINN seeds; all judged by the true Fourier model over the same feasible region).

method	fit RMSE (vol bp)	wall-clock (s)	Fourier evals
Fourier-loop from scratch	23.7	168	2018
PINN-only (single-shot)	222 ± 20	~ 2	0
Hybrid (PINN + Fourier polish)	29.4 ± 7.1	23	237 ± 46

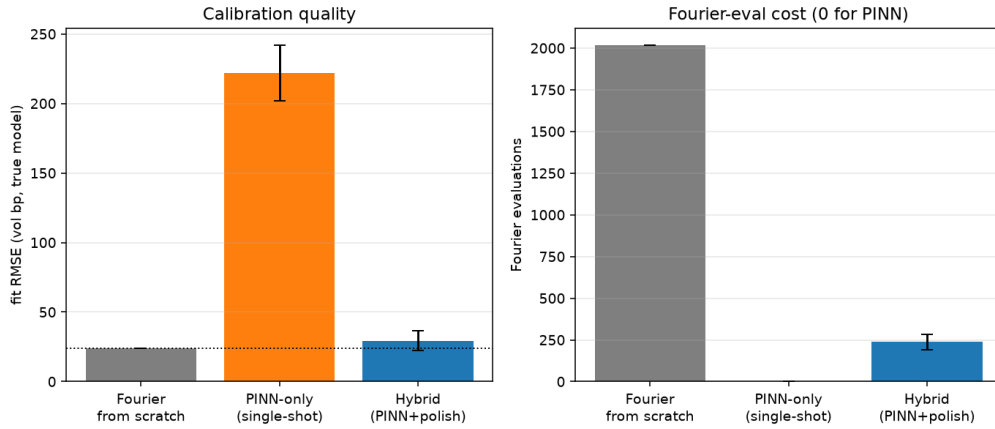


Figure 3: Hybrid calibration. Fit quality (left) and Fourier-evaluation cost (right) for the three methods on the BTC slice: the hybrid matches from-scratch quality at $\sim 9\times$ fewer Fourier evaluations, while PINN-only is fast but inaccurate.

5 Discussion

[Discussion — written last (Stage 4).]

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